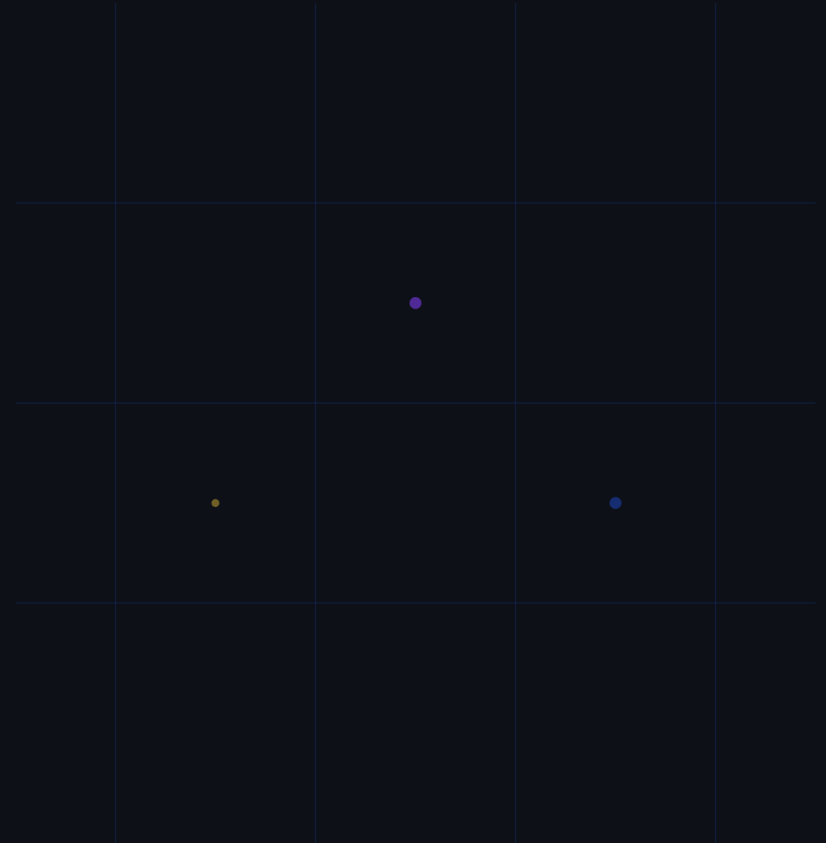


SEQUOIA CAPITAL

Generative AI's Act Two

From Technology Hammer to Real Value

Sonya Huang, Pat Grady and GPT-4



Six Decades of Technological Progress Converged to Enable Generative AI

TECHNOLOGICAL FOUNDATIONS



6 Decades of Moore's Law

Compute: delivered exaflops of processing power to train and run foundation models

1960 → Today



4 Decades of the Internet

Data: produced trillions of tokens of training data (accelerated by COVID)

1983 → Today



2 Decades of Mobile & Cloud

Distribution: a supercomputer in every human's palm

2003 → Today



GENERATIVE AI

Foundation Models

ChatGPT launch 2022

"Cambrian explosion"

"ChatGPT's launch was the spark that ignited a Cambrian explosion of innovation not seen since the early internet."

— Sonya Huang, Pat Grady & GPT-4, Sequoia Capital

GenAI Startups Surpassed \$1B Revenue Faster Than SaaS Ever Did

COMMERCIAL TRACTION

\$ >\$1B

Revenue from Startups Alone

Faster than SaaS took years to reach

vs. SaaS: measured in months, not years



#1

ChatGPT: Fastest-Growing App

In history — strong PMF with students

and developers worldwide



2 hrs

Character.ai Avg Session Time

Per user — AI entertainment leader

Popularized AI companionship category

Standout Performers

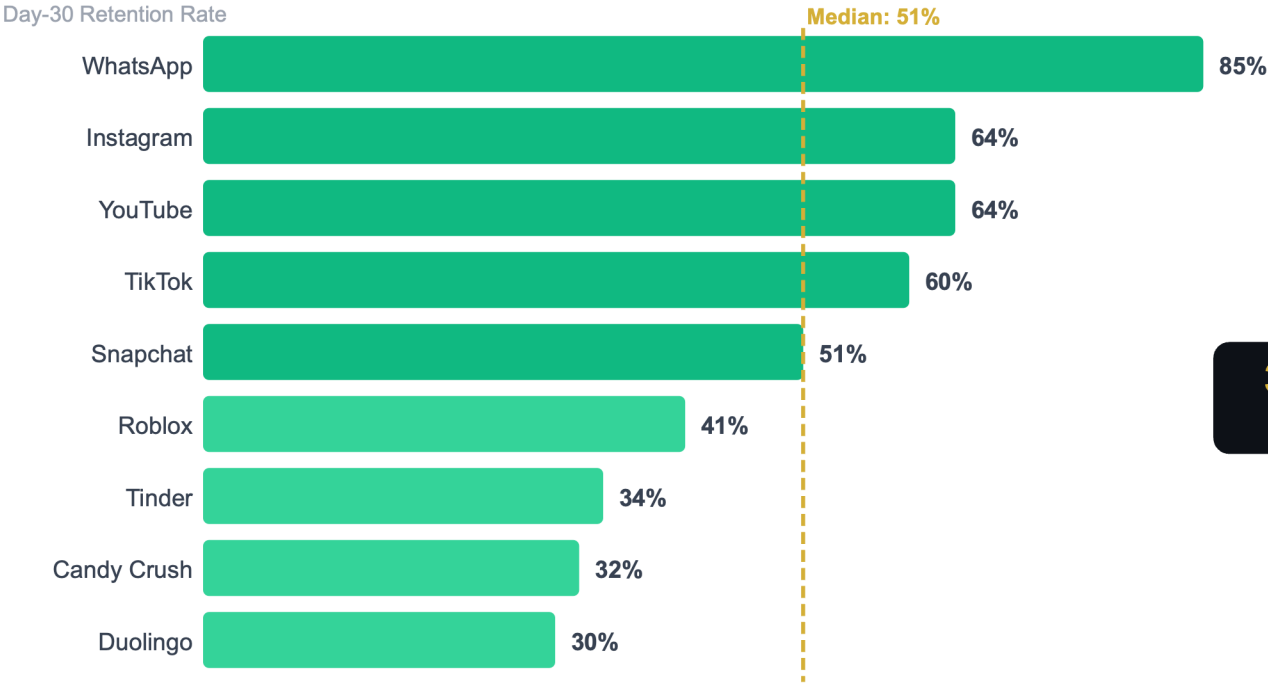
- **ChatGPT**
Fastest-growing application in history — strong product-market fit among students and developers
- **Midjourney**
Reported hundreds of millions of dollars in revenue with a team of just 11 people
- **Character.ai**
Popularized AI entertainment and companionship — users spend an average of 2 hours per session in-app

Critical caveat: These successes don't change the reality that many AI companies lack product-market fit or sustainable competitive advantage. The retention data tells a different story.

AI-First Apps Show 14% Median Retention vs. 51% for Incumbents — a 3.6× Gap

Established Consumer Apps

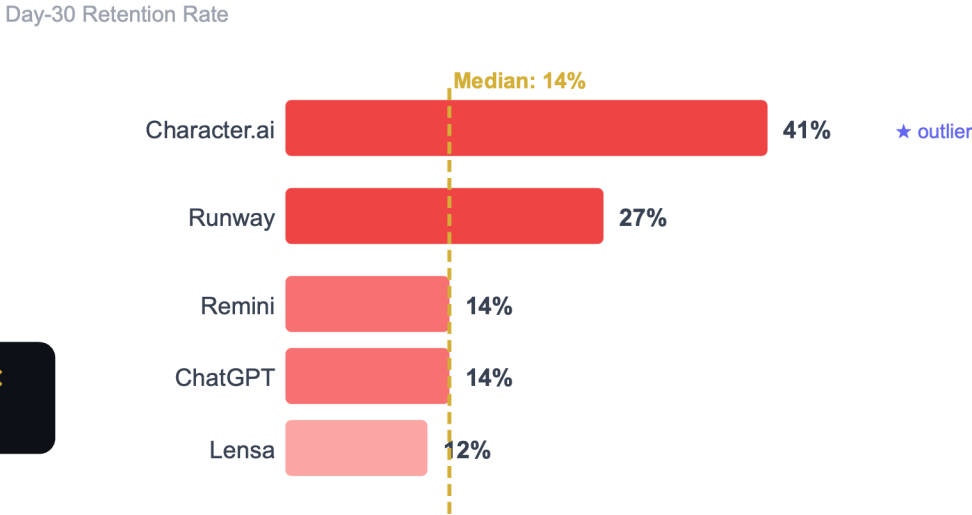
Day-30 Retention Rate



Incumbent Median: 51%

AI-First Companies

Day-30 Retention Rate



3.6×

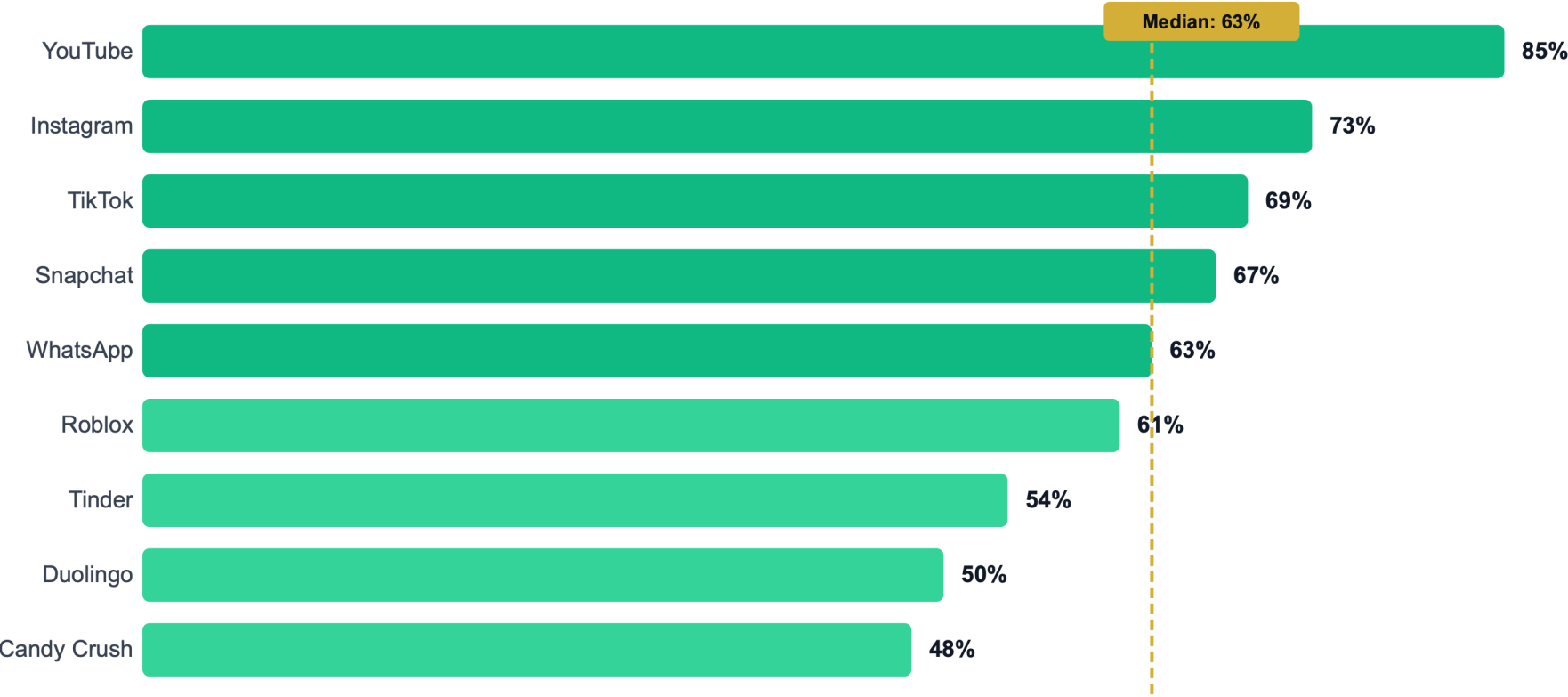
GAP

AI-First Median: 14% — 37 percentage points below incumbents

★ Character.ai (41%) is the AI-first retention outlier — companionship may be generative AI's strongest consumer

YouTube Leads Incumbents at 85% Monthly Active Usage — The Bar AI Must Clear

INCUMBENT ENGAGEMENT BENCHMARK | Monthly Active Usage Rate



The Bar AI Must Clear

Incumbent median of 63% monthly active usage dwarfs the AI-first 14% Day-30 retention median.
AI-first apps must close a 49-percentage-point gap to achieve comparable engagement.

ACT 1: Technology-Out

2022–2023 | Discovery Phase

- ❌ **Foundation model = entire product**
Models treated as complete solutions, not components
- ❌ **Novelty demos & thin wrappers**
"AI Salesforce", "AI Adobe" — undifferentiated copilots
- ❌ **No workflow integration**
Point solutions that don't embed into how people work
- ❌ **Poor retention — 14% median**
No durable moat; users don't return
- ❌ **Technology-first question**
"What can the model do?" — solution in search of a problem
- ❌ **Modality-based market structure**
Organized by text / image / code — not by customer need

EXAMPLES

Undifferentiated copilots · AI Salesforce · AI Adobe
Novelty chatbots · Lightweight demos

ACT 2: Customer-Back

2023+ | Value Delivery Phase

- ✅ **Foundation model = one component**
Integrated into comprehensive, multi-layer solutions
- ✅ **New editing interfaces**
Workflows become stickier; outputs become better
- ✅ **Multi-modal, end-to-end solutions**
Solves human problems completely, not partially
- ✅ **Durable retention via workflow lock-in**
Users return because the solution owns the workflow
- ✅ **Customer-first question**
"What does the user need end-to-end?" — problem-first
- ✅ **Use-case-based market structure**
Organized around human problems — legal, search, companions

ACT 2 EXEMPLARS

Harvey (legal) · **Glean** (enterprise search) · **Character**
Ava (digital companions)

The Sequoia market map is now organized by use case, not model modality — reflecting the Act 1 to Act 2 evolution

Market Map Reorganized by Use Case — Not Model Modality

FROM: Text / Image / Code modalities → TO: Human use cases and problem domains



Legal Harvey

Custom LLMs for elite law firms — full-stack legal workflow integration

Act 2 Exemplar



Enterprise Search Glean

Crawling and indexing workspaces for relevant enterprise AI search

Act 2 Exemplar



Companions & Social Character.ai · Ava

Digital companions — 2-hour avg session time; strongest consumer retention

Act 2 Exemplar · 41% retention



Creative & Visual Midjourney

\$100M+ revenue with 11 employees — image gen as creative workflow

Commercial breakout



Developer Tools ChatGPT (developer)

Fastest-growing app in history — strong PMF with developers globally

Multi-modal expansion



Enterprise Workflows Multi-modal category

Multimodal, workflow-integrated solutions crossing text, image, voice, code

Applications are increasingly multi-modal

The modality lens is obsolete.

The updated Sequoia market map is organized by use case, reflecting how customers actually experience value — not how models are built.

Five Principles for Building and Investing in Act Two

KEY TAKEAWAYS | Sequoia Capital Analysis

01 Retention is the critical signal

AI-first apps at 14% median vs. 51% for incumbents — a 3.6× gap that exposes lack of real utility

Character.ai (41%) is the outlier — companionship may be generative AI's strongest consumer use case

02 Foundation models are a component, not the product

Act 2 winners integrate models into comprehensive, workflow-embedded solutions — not thin wrappers

Harvey, Glean, Character, Ava are building around the model, not on top of it

03 Solve human problems end-to-end

Multi-modal, editing-interface-rich solutions that are stickier and harder to displace

New editing interfaces make workflows stickier and outputs better

04 \$1B+ revenue proves real demand exists

GenAI startups reached \$1B+ faster than SaaS — the opportunity is real despite the hype cycle

Midjourney: hundreds of millions in revenue with 11 employees

05 Use case beats modality

The new market map is organized around human problems: legal, enterprise search, companionship, creative, developer

Companies solving end-to-end human problems will define Act 2

Sonya Huang, Pat Grady and GPT-4 | Sequoia Capital | September 2023